Data Acquisition Unit (DAU) (Mod 6/2049) – General Data

1. General

The Data Acquisition Units (DAUs), DAU 1 and DAU 2 receive analog signals from the aircraft sensors and transmitters, and converts the analog signals to digital signals.

2. Description

A. Data Acquisition Unit (DAU)

The aircraft is equipped with two dual-channel Data Acquisition Units (DAUs) that collect and process various discrete, resistive, variable voltage and other signals from the engines and airframe and then encode this data into digital format for transmission via ARINC 429 to the MAU. The DAUs also do any logic required to trigger CAS messages. The DAUs are mounted above the cabin headliner, at approximately station 250. The DAUs are flange mount units and attach directly to the shelf.

The Apex ARINC 429 outputs to the MAU are a combination of graphics commands to drive systems indications, CAS message triggers, monitors that trigger aural warnings and PFD alerts, and phase of flight inhibit logic bits that are used by the MAU Monitor and Warning Function (MWF) to inhibit predefined CAS messages during certain phases of flight. The ARINC 429 data to the MAU also includes data required by the MAU to do logic related to avionics systems. The FDR ARINC 429 outputs to the MAU contain all applicable FDR data which the MAU converts to ARINC 717 and combines with other APEX FDR data before it is sent to the FDR (if installed). The FDR ARINC 429 outputs are also brought to the cross side DAU and provide cross side communication.

28 Volt DC power is provided from the left bus through 5 amp DAU 1A (E5) and DAU 2A (F5) circuit breakers, and from the right bus through 5 amp DAU 1B (E6), and DAU 2B (F6) circuit breakers on avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

1) The DAUs interface with the following aircraft systems and sensors:

a) Torque Transmitters

b) T_5 Sensors

c) Propeller RPM Transmitters

d) Gas Generator RPM Transmitters

e) Fuel Flow Transmitters

f) Oil Pressure Transmitters

g) Oil Temperature Transmitters

h) Generator Shunts

i) Battery Shunt

j) Brake Pressure Transmitter

k) Hydraulic Pressure Transmitter

l) Stall Warning Vane

m) Cockpit Temperature Sensor

n) Cabin Temperature Sensor

o) Elevator Trim Tab Potentiometer

p) Elevator Angle Position Potentiometer

q) Aileron Trim Tab Position Potentiometer

r) Aileron Position Potentiometer

s) Flap Position Potentiometer

t) Rudder Trim Tab Potentiometer

u) Rudder Angle Potentiometer

v) Aircraft Voltages via MAU ARINC 429

w) Aircraft Discretes via MAU ARINC 429

2) Some parameters were included as provisions. These parameters are jumpered, either directly or with drop down resistors, to other working parameters to prevent a DAU Maintenance CAS message. The provisions include:

a) Pilot Oxygen Transmitter (provision only)

b) Passenger Oxygen Transmitter (provision only)

c) Flap Lever Potentiometer

d) Engine Control Lever Potentiometers

e) Avionics Battery Voltage via the MAU ARINC 429

B. DAU Configuration Plugs

Two pin D Series DAU configuration plugs are used to define various aircraft parameters. They are located on either side of the DAU maintenance panel, above the cabin headliner, just forward of the DAUs at approximately station 244.00.

Parameters controlled by the configuration plugs include:

1) Flaps 20 permitted (provision only)

2) Engine Type

3) De-ice Installed

4) Pitot Heat Warning

5) Oxygen Installed (provision only)

6) Fuel in Kgs (provision only)

7) Gear Configuration

3. Adjustment/Test

A. Data Acquisition Unit (DAU)

If a fault condition is detected by the Built-In Test Equipment (BITE), a caution level CAS message will be displayed.

1) Ensure no DAU Fail or DAU Maintenance CAS messages are present following system initialization.

2) Ensure the following system windows are displayed properly:

a) Engine Window

b) CAS Window

c) Flight Controls/Hydraulic Window

d) Fuel Window

e) Electrical/Misc. Window

f) Environmental Window

4. Removal and Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT/REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MOUNTING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDING – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Data Acquisition Unit (DAU)

1) General

These procedures are for the removal and installation of DAU 1 and DAU 2.

2) Removal of Data Acquisition Unit (DAU)

a) Ensure the aircraft electrical system is OFF.

b) Open the DAU 1A (E5), DAU 2A (F5) (Left Bus), DAU 1B (E6) and DAU 2B (F6) (Right Bus) circuit breakers and fit safety clips.

c) Gain access to the ceiling compartment aft of the bulkhead 240.

d) Disconnect (4) electrical connectors.

e) While supporting the DAU, remove (4) mounting screws.

f) Remove unit from shelf.

Note Stow connectors of the removed component.

3) Installation of Data Acquisition Unit (DAU)

a) Ensure the aircraft electrical system is OFF.

b) Ensure DAU 1A (E5), DAU 2A (F5) (Left Bus), DAU 1B (E6) and DAU 2B (F6) (Right Bus) circuit breakers on the avionics circuit breaker panel are open and fitted with safety clips.

c) Lift DAU into position and install (4) mounting screws.

d) Reconnect (4) electrical connectors. Ensure connectors are tight.

Note It is possible to swap connectors P2 and P3. It is also possible to swap P4 and P5. Ensure connectors are installed in the correct positions to prevent erroneous operation or damage.

e) Install interior as required.

f) Remove safety clips and close the DAU 1A (E5), DAU 2A (F5) (Left Bus), DAU 1B (E6) and DAU 2B (F6) (Right Bus) circuit breakers.

B. DAU Configuration Plugs Change

Note The DAU configuration plug must match current gear configuration. VSP18–XXXX. The VSP18–XXXX part number is unique to each aircraft's configuration options. Contact Viking Technical Support for configuration option.

1) Ensure the aircraft electrical system is OFF.

2) Open the DAU 1A (E5), DAU 2A (F5) (left bus), DAU 1B (E6), DAU 2B (F6) (Right Bus) circuit breakers.

3) Gain access to the ceiling compartments aft of bulkhead 240.

4) Change the DAU configuration plugs to match the gear configuration located of FWD DAU mounting plate left and right hand side.

5) Close the DAU 1A (E5), DAU 2A (F5) (left bus), DAU 1B (E6), DAU 2B (F6) (Right Bus) circuit breakers.

5. Troubleshooting

A. Data Acquisition Unit (DAU)

1) Ensure DAU 1A (E5), DAU 2A (F5) (Left Bus), DAU 1B (E6) and DAU 2B (F6) (Right Bus) circuit breakers are closed.

2) Visually inspect the electrical connectors for damaged wiring or contacts. Ensure connectors are fully seated and tight.

3) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

4) If repair or service is required, send to appropriately rated avionics repair facility.